Design of primary and composite routing metrics for RPL-compliant Wireless Sensor Networks

Panagiotis Karkazis
Dept. of Electronic & Computer Engineering,
Technical University of Crete,
Chania, Greece
{pkarkazis@isc.tuc.gr}

Helen C. Leligou, Lambros Sarakis, Theodore Zahariadis
Dept. of Electrical Engineering,
TEI of Chalkis, Psahna, Greece
{leligou, sarakis, zahariad}@teihal.gr

Terpsi Velivassaki, Christos Capsalis
Dept. of Electrical Engineering
National Technical Univ. of Athens
Athens, Greece
tveli@mail.ntua.gr, ccaps@central.ntua.gr

Abstract—**The diversity of applications that current and emerging Wireless Sensor Networks are called to support imposes different requirements on the underlying network with respect to delay and loss while at the same time the WSN imposes its own intricacies. The satisfaction of these requirements highly depends on the routing metric upon which the routes are decided. In this view, the ROLL group in IETF has proposed the RPL routing protocol, which can flexibly work on various routing metrics as soon as they hold specific properties. The system implementer/user is free to decide whether to use one or multiple routing metric as well as the way these metrics can be quantified and combined. In this paper, we provide ways to quantify the routing metrics so that they can be combined in additive or lexical manner. We use extensive simulation results to evaluate the impact of the routing metrics on the achieved performance.**

Keywords-component; Wireless sensor networks, composite routing metrics, QoS differentiation, IETF ROLL, RPL

I. INTRODUCTION

The main goal of any routing protocol is to support effective communications. To achieve it, the design of a routing protocol must be based on the characteristics of its target networks and applications. For example, the mobility of nodes in ad-hoc networks demands routing protocols that can converge rapidly and maintain connectivity in an efficient manner, while the severe energy constraints of wireless sensor networks demand the design of energy-efficient routing and heavy traffic load in mesh networks requires load-balanced routing schemes. Regarding the target applications, simple condition control (humidity, temperature) can be considered as delay tolerant application in contrast to military and border control applications which impose strict latency and reliability requirements. The design of effective routing metrics, however, is not a trivial task. Existing literature focuses on designing different routing metrics for different wireless networks [1]. However, there is a lack of knowledge about the impact of these routing metric designs on the overall operation of routing protocols. In other words, the routing metrics cannot be arbitrarily exchanged (one routing metric to another) without considering the routing protocols used in each specific network. Given that Wireless Sensor Networks (WSNs) consist of resource-constrained devices and are used in a variety of applications, to optimize performance, it is necessary to design proper composite metrics.

The growing interest in WSNs has led IETF to the establishment of specific groups working on protocol design for Low power Lossy Link Networks (LLNs) and among them, the Routing Over Low power and Lossy networks (ROLL) group. This group has standardized the so-called IPv6 Routing Protocol for LLNs (RPL) [9] which provides a mechanism whereby multipoint-to-point traffic from devices inside the LLN is routed towards a central control point, as well as point-to multipoint traffic from the central control point to the devices inside the LLN. RPL constructs Directed Acyclic Graphs (DAG) and defines the rules based on which every node selects a neighbor as its parent in the DAG, thus forming a graph. To cover the diverse requirements imposed by different applications, ROLL has specified in [10] a set of link and node routing metrics and constraints (which can be static or dynamic) suitable to LLNs to be used in RPL. This document does not provide details on the quantification of each routing metric, leaving it to the implementer to decide how to express each metric; it only provides examples. Moreover, while it is possible to combine multiple routing metrics, no guidelines are provided. Up to now, ROLL has specified in [11] only the Objective Function 0, (OF0) where the hop count is the only routing metric adopted.

In this paper, we propose formulas for the evaluation of the RPL routing metrics and we enrich the routing metric set by a new routing metric which targets the detection and avoidance of malicious/malfunctioning nodes. Moreover, placing emphasis on the need for composite routing metrics, we propose the use of additive and lexical metric composition and we investigate the performance of the RPL protocol for different (primary or composite) routing metrics using computer simulations to reach both quantitative and qualitative results.

The rest of the paper is organized as follows: in section II, the RPL protocol rules and operation are outlined, so that proper routing metrics are designed in section III. Section IV includes the simulation results and conclusions are drawn in section V.

II. APPLICABILITY TO RPL

For the construction of the graph (DAG) in RPL each node calculates a rank value and advertises it in the so-called

Destination Information Object (DIO) messages. The rank of a node is a scalar representation of the location of that node within a destination-oriented DAG (DODAG). In particular, the rank of the nodes must monotonically decrease as the DODAG version is followed towards the DAG destination (root). Each node selects as parent the neighbor that advertises the minimum rank value. This guarantees loop free operation and convergence [2]. The rank has properties of its own that are not necessarily those of all metrics. On the path from the destination towards the source nodes, the rank is incremented in a strictly monotonic fashion and can be used to validate a progression from the root towards the leaf nodes while a metric, like bandwidth or jitter, does not necessarily exhibit this property. It is the Objective Function (OF) that defines how routing metrics are used to compute the rank, and as such it must demonstrate certain properties. As an example, adopting OF0 (which is the only objective function defined so far) each node announces how many hops away from the destination/root it is located. In this case, the rank reflects the node's proximity to the root of the DAG. The OF dictates how parents in the DAG are selected and thus, it affects the DAG formation.

Transferring the formalism explained in [2], [3], a node's metric value, is equivalent to the node's Rank, and reflects the node and link characteristics formalism that will lead the parent selection process. In RPL,

$$Rank_i = Rank_j + w_{i,j}$$

where node j is node's i potential parent and $w_{i,j}$ is a function of the characteristics of node j and of the link between nodes i and j. This can be expressed as

$$\omega(a \oplus b) = \omega(\alpha) + \omega(\beta)$$

using the formalism of Sobrinho, where α is the path from the root node to node j and β is the link between node i and j. According to RPL specifications, node i will select as parent, the node that minimizes the Rank value among all potential parents. For example, given that

$$Rank_i = Rank_j + w_{i,j}$$
$$Rank_i' = Rank_k + w_{i,k}$$
$$Rank_i < Rank_i'$$

then node i will choose node j as its parent, since

$$Rank_i = \min\{Rank_i, Rank'_i\}$$

Since the rank should monotonically increase, $w_{i,j}$ on the path from the root to the leaves (equivalently $\omega(a)$) must be a positive (non zero) number, whose range is bounded by a minHopRankIncrease and a maxRankIncrease value, respectively. According to RPL specifications, minHopRankIncrease value is defined as the minimum increase in Rank between a node and any of its DAG parents, while maxRankIncrease is defined as the maximum value increase that a given node can advertise within the same DAG version.

III. Designing Routing Metrics for WSNs

An important number of routing metrics have been proposed for WSNs to capture different network and node characteristics and for some of them an even greater number of ways to quantify them has been presented. Among them we find metrics indicating the distance from the destination measured in hops (hop count) or in physical distance units (measured on physical or virtual coordinates), the packet loss due to congestion or link failures measured as the number of expected transmissions until the packet is successfully transmitted (expected transmissions number –ETX), the remaining energy of a node or of the nodes in a path. Different ways to quantify them exists (as discussed in [10]) with fifteen ways to quantify ETX proposed in [6] and a similar number reported in [7] for energy.

The ROLL group has specified in [10] the node energy (NE), hop count (HC), link throughput, Link Latency, Link Quality Level and Expected Transmission Count (ETX) routing metrics while leaving space for defining additional primary routing metrics. We now seek ways to quantify them satisfying the previously mentioned property and allowing for easy (straightforward) combinations into composite routing metrics to meet diverse performance criteria.

Our aim is to design primary routing metrics which capture WSN relevant node, network characteristics and which are proved to be monotonic and are suitable for building composite metrics to meet diverse application requirements, still satisfying the monotonicity requirement of the RPL protocol.

To rigorously describe a routing metric, a wireless network is defined as a strongly connected directed graph *G(V, E)*, where *V* is the set of nodes and *E* the set of edges representing wireless links between neighboring nodes. The design of a routing metric can be formally defined as an algebra on top of a quadruplet $(S, \oplus, \omega, \leq)$, where S is the set of link weights, ω is a function that maps a path or a link to a weight, $\leq$ is an order relation and $\oplus$ is the path concatenation operation. This quadruplet is named *path weight structure* and it is the mathematical representation of a routing metric.

A. Primary routing metrics

A routing metric that has been widely used to decide the cost of the routes in WSN is the ***hop count***. It is a primary routing metric that is used to report the number of traversed nodes along the path. It can be represented as $(S, \oplus, \omega, \prec)$, where S is the set of network edges, the weight of a path is equal to the sum of weights of the links which produce the path when concatenated i.e. $\omega(a \oplus c) = \omega(a) + \omega(c)$, for any paths a, c, the weight of the root/sink node is equal to 0, the weight values range in the positive integers, and $\prec$ is the "less than" order defined for integers. This metric increases strictly monotonically from the root towards the leaf nodes which completely justifies the design of OF0.

The **link latency** is a primary routing metric that increases strictly monotonically from the root towards the leaf nodes, is always a positive real number and is minimizable, i.e. the better path is the path with the lower link latency. Although the latency is more a link than a node metric, it exhibits the same monotonicity properties as hop count.

Minimizing the traversed nodes (i.e. the hop count), the overall number of transmissions is expected to be minimized which leads to the consideration that also the overall energy consumption and the packet latency are reduced. Unfortunately, this is true only under the assumption of equally loaded and equally lossy links which in general does not hold.

To discriminate lossy and/or congested paths, **the Expected Transmission Count** (ETX) has been proposed and defined as the number of transmissions (including retransmissions) a node expects to make towards a destination in order to successfully deliver a packet, according to [10]. The ETX of a path is defined as the summation of the ETX's of all links along the path (see [5]) and on each link it expresses the number of link layer transmissions required for the successful delivery of a message to the next hop neighbour. The successful delivery of a link layer message is decided based on the reception of link layer acknowledgment. Based on ETX the lossy links are distinguished irrespective of the cause of loss, e.g. physical layer causes or contention.

Assuming that node i transmits towards node j, realizing link α and that s packets were successfully delivered and f failures were observed (through the absence of link layer acknowledgement), the related metric can be quantified as

$$ETX^{i,j} = \frac{s+f}{s} \geq 1$$

$ETX^{i,j}$ is a positive real number, it is monotonically increasing (considering this as an additive metric along the path) and is minimized to the lower advertised values (i.e. the node advertising the lowest value is preferred). For the routing module of any node to maintain the ETX-related information, the routing layer has to communicate with the link layer, mandating the realization of a cross-layer interface.

Another link-reliability metric (also identified in [10]) is the **Link Quality Level**. The Link Quality Level takes values in a limited set of 7 levels with 1 reporting the highest link quality level and the LQL of a path is expressed as a record containing for each encountered LQL value, only the number of matching links. If a user adopts the example rule mentioned in [10], defining that each node selects the path with most links reporting a LQL value of 3 or less, then the respective metric is not strictly monotonic. Concatenating any path α with a path b consisting of links with LQL higher than 3, the concatenated path $(a \oplus b)$ will have $\omega(a \oplus b) = \omega(a)$. To render the related metric strictly monotonic, LQL can be defined as

$$\omega(\alpha) = \sum_{k=1}^{7} p_k k$$

where k is the link quality level and p_k the number of links matching the k-th LQL. Then $\omega(a)$ is strictly monotonic and minimizable, i.e. the lower $\omega(a)$ reveals a better path. However, this approach mandates the definition of quality levels in order to classify each link to one of the 7 supported levels. Additionally, as level 0 corresponds to links with unknown quality, paths traversing such links should be avoided since it makes no sense to take into account the LQL of path if at least one link has unknown quality level.

In energy-aware routing, the routing decisions depend on considerations of the available energy of the nodes. These considerations can be significantly more complicated than simply finding the route with the lowest hops. In fact, the shortest (in terms of hops) path seems to introduce the lowest energy consumption, which is not necessarily true, if we take into account the retransmission that may be needed. If the remaining energy is not taken into account, then few paths can be more loaded than others and nodes closer to the sink are more subject to premature energy depletion, since they have to relay more packets, which can eventually lead to network disconnection.

To discriminate paths traversing "weak" (residual energy-wise) nodes, any direct link α is scored by the metric $\omega(\alpha)$ which reflects the remaining energy of the end node i of the direct link α (expressed as the ratio between the maximum (initial) energy $V_{\max}$ and the current energy value V_{now},), i.e.

$$\omega(\alpha) = RE^i = \frac{V^i{}_{\max}}{V^i{}_{now}} \geq 1$$

then the energy metric $(S, \oplus, \omega, \prec)$ can be defined as $\omega(\alpha \oplus b) = \omega(\alpha) + \omega(b) > 1$ since both ω(α) and ω(b) are both greater than or equal and $\prec$ the "less than" relation over real numbers. Based on this metric, the path traversing nodes with higher energy will be selected, among paths of the same length and the path traversing less nodes will be preferred among paths passing over nodes with similar remaining energy levels.

In WSNs, apart from layer 2 loss, layer 3 losses may occur: a very common routing attack ([8]) in WSNs is the black hole / grey hole attack during which a node refuses forwarding all / part of the traffic acting either selfishly (to economise energy) or maliciously, even though it acknowledges the reception of the traffic at layer 2. The impact on the network performance is aggravated if it additionally advertises "good" routes to the root, where "good" is expressed according to the adopted routing metric. To account for this cause of loss (which we consider as routing layer losses), we consider a trust-related metric which is evaluated as follows: each node after transmitting a packet to a neighbor, it enters the promiscuous mode and waits to listen whether the selected neighbor has actually forwarded its packet, as proposed in [12],[13],[14], thus building trust knowledge. Assuming that sf packets were actually (successfully) forwarded and ff packets failed to be forwarded (realized through the absence of overheard packet being forwarded), the mean value of the probability for a packet to successfully travel along this path (without being dropped) is (as also reported in [15]):

$$P_{succ} = \prod_{i=1}^{m} \frac{sf_i}{sf_i + ff_i}$$

Where i ranges over the m links that form the path from the source to the destination. Equivalently, the "best" path is the one that corresponds to the lowest value of the ratio

$$\frac{1}{P_{succ}} = \prod_{1}^{m} \frac{sf_i + ff_i}{sf_i}$$

Since the logarithmic function is monotonically increasing, to compare any two paths based on 1/Psucc, the $\log(\frac{1}{P_{succ}})$ can be equivalently compared.

We define the packet forwarding indication (PFI) metric $(S, \oplus, \omega, \prec)$ for any link α as,

$$\omega(\alpha) = \log(\frac{1}{P_{succ(a)}}) = \log(\frac{sf_a + ff_a}{sf_a}) \geq 0 \quad (6)$$

and $\prec$ the "less than" relation over the (positive) real numbers.

For any path produced by the concatenation of paths α and b, $\omega(\alpha \oplus b) = \omega(\alpha) + \omega(b)$. The PFI metric is monotonic (although not strictly) and for this reason it has to be combined with a strictly monotonic primary routing metric to be suitable for use in RPL.

B. *Designing Composite routing metrics*

The combination of multiple primary routing metrics into a composite one can lead to the optimization of more than one performance aspect. However, the monotonicity property of the composite routing metric has to hold for the routing protocol to be loop-free. Gouda et. Al. have defined in [4] the lexical and additive routing metric compositions which have been used to decide the tree structure in this work.

The additive composition relation $\prec_{add}$ over the set $S_1 \times S_2$ is defined as follows:

$$(\omega_1(\alpha),\omega_2(\alpha)) \prec_{add} (\omega_1(b),\omega_2(b)) \Leftrightarrow \omega_1(\alpha)+\omega_2(\alpha) < \omega_1(b)+\omega_2(b)$$

The two primary metrics should hold the same order relation ($\prec$ or $\succ$) so that the produced composite additive routing metric makes sense. It is obvious that the strict monotonicity property of the additive routing metric holds for any combination of primary routing metrics that are strictly monotonic or for combinations of strictly monotonic with monotonic. Thus, for example, the PFI metric has to be combined with any primary routing metric among Hop count or ETX. It is worth stressing that the monotonicity for example of the ETX metric depends on its definition and does not hold for any of them as proved in [5].

We generalize the definition of the additive composite routing metric and define that the weight of a path can be expressed as

$$\omega(\alpha) = a_1\omega_1(\alpha) + a_2\omega_2(\alpha)$$

where (a_1, a_2) is a pair of positive real numbers which represent the relative weights of the two metrics and enable the shift of emphasis between the two primary routing metrics as will be shown using computer simulations.

Another way to combine primary routing metrics (also defined in [4]) is the lexical combination of primary routing metrics, which provides a clear prioritization of the metrics since they are inspected in strict priority.

The lexical metric composition of two routing metrics is defined as follows: Consider two routing metrics $(S_1, \oplus_1, \omega_1, \prec_1)$ and $(S_2, \oplus_2, \omega_2, \prec_2)$, where $\prec_1$ is the order relation over a set of metric values S_1 and $\prec_2$ is the order relation over a set of metric values S_2. Then, the relation $\prec_{lex}$ over the set $S_1 \times S_2$ is defined as the lexical composition over the ordered pair $\langle \prec_1, \prec_2 \rangle$ if and only if for every link pair α , b, which are mapped to the weight pairs $(\omega_1(\alpha), \omega_2(\alpha))$ and $(\omega_1(b), \omega_2(b))$ in $S_1 \times S_2$:

$$(\omega_1(\alpha),\omega_2(\alpha)) \prec_{lex} (\omega_1(b),\omega_2(b)) \Leftrightarrow$$
$$(\omega_1(\alpha) < \omega_1(b)) \vee \left(\omega_1(\alpha) = \omega_1(b) \wedge \omega_2(\alpha) \prec_2 \omega_2(b)\right)$$

An interesting feature of the lexical combination is that it is not necessary for the primary metrics to hold the same order relation, since the comparison takes place among values corresponding to the same metric, i.e. the value corresponding to ω_1 for all path are ordered first using the same relation order $\alpha_1$and then among those paths with the same weight ω_1 the value for ω_2 are considered for ordering the paths using the relation α_2.

With respect to the monotonicity, according to [9], to guarantee the strict monotonicity of the lexical composite routing metric, the first of any two selected metrics has to be strictly monotonic. Given that from the metrics discussed in the previous section, Hop Count, ETX and Remaining energy are strictly monotonic, any lexical combination of them is also strictly monotonic.

IV. SIMULATION RESULTS

The JSIM [16] open simulation platform has been used to model RPL [17]. The topology tested consists of 100 nodes placed on a 10x10 grid, where multiple nodes were generating data messages towards the same destination (root node). The activated data sessions periodically generate data. Apart from tools to measure performance in terms of packet latency and the packet loss, we also developed a graphical user interface which shows the constructed paths in real-time, thus allowing us to observe all dynamic path alterations. During our investigation, the routing metrics tested included Hop Count, ETX, PFI and RE either as single routing metrics or in combination with each other.

A. Combining HC with PFI

If only HC is considered, the shortest path to the root is defined and each node selects as its parent the one-hop neighbor that is closer to the root. To create short paths avoiding misbehaving nodes (acting selfishly or maliciously), the hop count can be combined with PFI. To understand how the composite routing metrics work and evaluate the performance benefits, we have run scenarios combining the hop count and Packet forwarding indication metrics for different penetrations of misbehaving nodes randomly distributed in the grid. The tested routing metrics include a)hop count (HC), (as the simplest routing metric used also for comparison reasons), b) the lexical combination of HC and PFI (marked as lex(HC, PFI) in the figures) where the hop count is inspected first and only if two paths have equal hop count metric then the PFI metric value is inspected and c) their additive combination (marked as Add(HC, PFI)) for different weight (α_1, α_2) pair values to show the impact of the these parameters on the achieved performance. The misbehaving nodes perform grey hole attacks dropping randomly half of the received traffic. Six data sessions are initiated with the sources being randomly distributed in the different runs and the root node being the node in the upper left corner of the grid topology.

The results regarding the packet loss are depicted in Fig. 1 and clearly show that if the hop count is the only routing criterion, then, even with low penetration of misbehaving nodes and even if these nodes perform grey hole attacks (dropping 50% of the received traffic and not all the received traffic as would happen in the case of black-hole attack), the packet loss raises very rapidly. This is alleviated for all tested composite routing functions combining PFI with hop count which offer better performance in terms of packet loss than HC when used on its own.

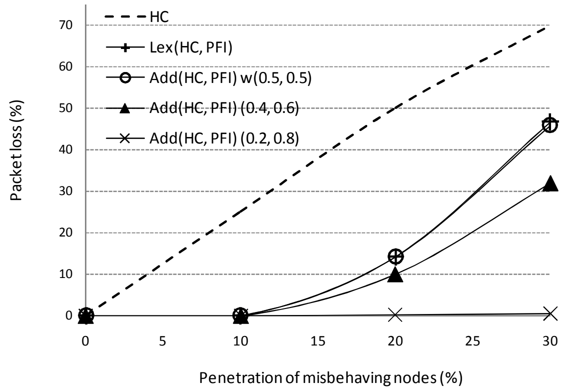

Figure 1. Performance comparison in terms of packet loss for routing metrics combining HC and PFI vs. the penetration of misbehaving nodes.

The improvement ranges from 5-68% which shows that the composite routing functions enable the detection of misbehaving nodes and the selection of paths that are more reliable. Looking at the composite routing metrics, the lexical combination of the hop count and PFI offers better performance than HC but worse than the considered additive combination of HC and PFI. This is due to the fact that in the case of lexical combination, the PFI metric is inspected only when alternative paths of equal length with different PFI values exist. For the additive combinations, the performance significantly depends on the metric weight pair (α1, α2) especially when the penetration of misbehaving nodes increases. When α1 is higher, the performance comes closer to that of the Lex(HC, PFI) while for lower α1 values the emphasis is shifted to PFI and lower packet loss is achieved.

The average latency results are included in Fig. 2 and show that for zero misbehaving nodes in the network, all the routing metrics tested lead to equal average latency. Comparing the latency observed for the HC with any composite routing metric, the prevalent observation is that the metric that leads to the best performance with respect to packet loss, leads to increased latency. This is due to the fact that, to avoid the misbehaving nodes, longer paths are usually followed in the case of composite HC, PFI metrics. It is worth stressing that the Lex(HC, PFI) results in latency performance very close to the one observed for the HC, while still offering a loss improvement of 23% for 30% malicious nodes in the network.

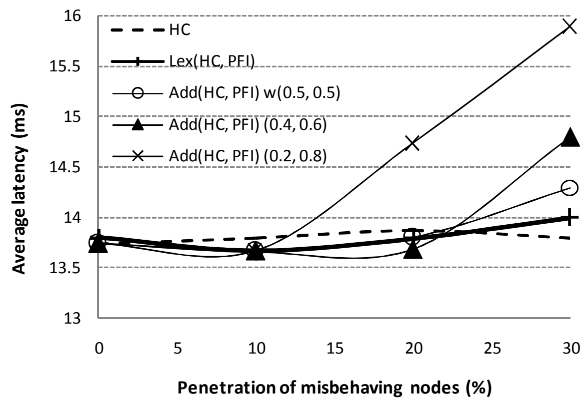

Figure 2. Performance comparison in terms of latency for routing metrics combining Hop Count and PFI vs. the penetration of misbehaving nodes.

B. Combining Hop count with remaining energy

In this section we investigate the combination of hop count with remaining energy (RE) considering that HC serves as a tool for reduced latency (a possible application requirement) and RE aims to satisfy a network requirement (for prolonged lifetime). We have carried out a simulation scenario set where several nodes are transmitting data packets towards the same destination-sink. When the Hop Count is the only routing metric used, the paths to the root are static, since no dynamic routing metric is taken into account, while they change dynamically when RE is also considered during routing decision making.

To investigate this effect, we have established six data sessions. When only the hop count metric is used, five of them pass through node 11 (one hop neighbour of the sink node) and one from node 12 (also one hop neighbour of the sink node) and one from node 68 which is not close to the sink node. The results regarding the energy consumption rate for these nodes, expressed in ratio of initial energy consumed every ms are depicted in Fig. 3. It is evident that when routing is decided based only on hop count, node 11 (which participates in five data sessions) has higher energy consumption rate than node 12 and 68 (which participate in one data session). Between node 12 and 68, node 12 (located closer to the root node) has higher energy consumption rate because it forwards the control messages of more nodes than node 68 located which is closer to the periphery of the grid network.

Combining hop count with the remaining energy either in a lexical or in an additive manner, the energy consumption of all nodes converges since the forwarding load is distributed among all one hop neighbours of the destination-sink node. The neighbors of node 11 and 12 will very quickly realize that node's 11 remaining energy has dropped and will select another node offering a path to the root. This is proven by our simulation results, which show that the energy consumption of node 11 for any composite HC, RE metric is lower than the one observed when the HC is the only metric for deciding the route. A lower difference is observed for node 12 since this forwards data from just one flow and the improvement potential is lower. Comparing the different composite routing metrics, the best load balancing is achieved when the additive routing metric places emphasis on the remaining energy. (In the figure, only the weight of the Hop Count is mentioned and the weight of the RE is given by $\alpha_1+\alpha_2=1$.) Any composite routing metrics brings an improvement ranging between 14 and 16% for node 11 and between 6-8% for node 12. Comparing the different composite routing metrics and with regards to the additive composite metric, we have tested different weight factors for hop count and remaining energy respectively (as indicated in the figure).

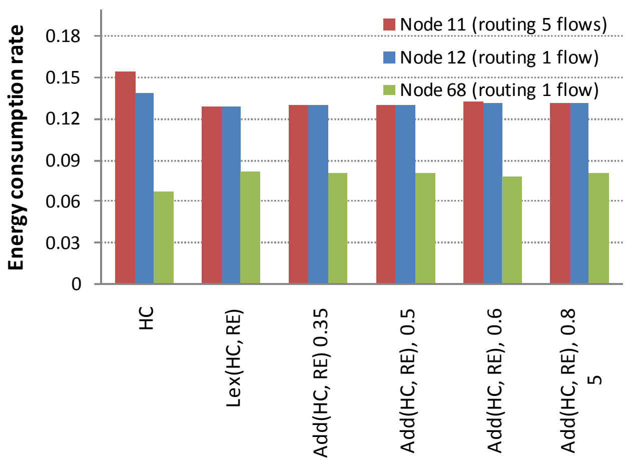

Figure 3. Performance comparison in terms of energy consumption rate expressed in ratio of initial energy consumed every ms combining HC and RE

V. CONCLUSIONS

In this paper we have focused on Low power Lossy link Networks and we have proposed formulas for quantifying primary routing metrics which capture effect relevant to the considered network type and we investigated their combinations. Combining primary metrics in a lexical manner leads to strict performance metric prioritization and can thus be used to ensure the application's requirements while optimizing other performance aspects to the extent possible. The additive composite routing metric on the other hand offers a more flexible way of combining metrics using the metric weight pair.

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